

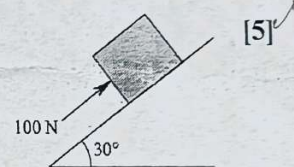
National Institute of Technology, Kurukshetra (HR)

B. Tech. Mid-Sem 2, 2025

Branch and Sem	: R & A, 2 nd	Time	: 50 Mins
Course	: Engineering Mechanics (RAPC-101)	Max. Marks	: 15
Session & Time	: Evening, 03:00-03:50 PM	Date	: 08/04/2025

NOTE: Attempt all the questions. Assume any missing data.

- Q1. A block of 50 kg mass is pushed up an inclined plane by a force of 100 N acting parallel to the incline as shown in figure. Determine whether the block shown is in equilibrium or not. Also, find the magnitude and direction of the frictional force. The coefficients of friction between the block and the plane are $\mu_s = 0.25$ and $\mu_k = 0.2$.



- Q2 In a shooting practice, a bullet is to be fired at a target, 150 m away at the same level. If the release velocity of the bullet is 60 m/s, determine what should be the angle of inclination of the gun with respect to the horizontal

- Q3. Planes A and B fly at the same elevation and have the motions shown in figure. Determine the velocity and acceleration of A as measured by the pilot of B.

